

**VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



**ASSIGNMENT
LOGIC DESIGN WITH HDL**

MATRIX MULTIPLICATION

MAJOR: COMPUTER ENGINEERING

HO CHI MINH CITY, 1/2025

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INTRODUCTION

1.1. AIMS

- Practice, improve the ability of using Verilog HDL to design digital circuits.
- Practice the analysis, design skills following hierarchy approach.
- Knowledge about popular circuits that are used in digital systems.
- Practice research, self-study, documentation, presentation skills.

1.2. REQUIREMENTS

- Work in a group of 4 to 6 members.
- Make demo scenarios. May design more modules to demo on FPGA if possible.
- Report structure:
 - Introduction: introduce the topic, scope and summary the report content.
 - Backgrounds and applications: theory of any concept in the topic and application of the circuit.
 - Design: present the ideal, block diagram, flow chart and algorithms of the design.
 - Implementation: technical document implemented blocks, modules.

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- Results: present the experiment setups, waveform, explanations and report timing, resources.
 - Conclusion: summary working achievements, advantages, disadvantages and future works.
- Compress the report (.pdf), source code (.v) in only 1 .zip file. Name the .zip file the group ID. Submit on LMS by deadline.

1.3. EVALUATION CRITERIA

- The content of presented theories, applications.
- The basic functions of implemented circuit.
- The content and presentation of the report.
- Source code: hardware design rule, coding rule, coding style.
- Resource utilization: number of LUTs, number of registers, power consumption and processing time.

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TOPIC

2.1. MATRIX MULTIPLICATION

In mathematics, specifically in linear algebra, **matrix multiplication** is a binary operation that produces a matrix from two matrices. For matrix multiplication, the number of columns in the first matrix must be equal to the number of rows in the second matrix. The resulting matrix, known as the **matrix product**, has the number of rows of the first and the number of columns of the second matrix. The product of matrices **A** and **B** is denoted as **AB**.

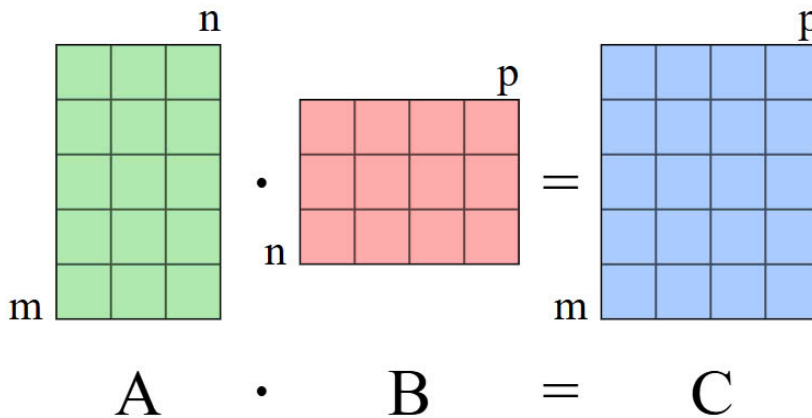


Figure 2.1: Matrix multiplication

Figure 2.2 shows the general formula for multiplying two square matrices of size 4x4.

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$$\begin{bmatrix} ax_1 & ax_2 & ax_3 & ax_4 \\ ay_1 & ay_2 & ay_3 & ay_4 \\ az_1 & az_2 & az_3 & az_4 \\ av_1 & av_2 & av_3 & av_4 \end{bmatrix} \times \begin{bmatrix} bx_1 & bx_2 & bx_3 & bx_4 \\ by_1 & by_2 & by_3 & by_4 \\ bz_1 & bz_2 & bz_3 & bz_4 \\ bw_1 & bw_2 & bw_3 & bw_4 \end{bmatrix} = \begin{bmatrix} ax_1*bx_1+ax_2*by_1+ax_3*bz_1+ax_4*bw_1 & ax_1*bx_2+ax_2*by_2+ax_3*bz_2+ax_4*bw_2 & ax_1*bx_3+ax_2*by_3+ax_3*bz_3+ax_4*bw_3 & ax_1*bx_4+ax_2*by_4+ax_3*bz_4+ax_4*bw_4 \\ ay_1*bx_1+ay_2*by_1+ay_3*bz_1+ay_4*bw_1 & ay_1*bx_2+ay_2*by_2+ay_3*bz_2+ay_4*bw_2 & ay_1*bx_3+ay_2*by_3+ay_3*bz_3+ay_4*bw_3 & ay_1*bx_4+ay_2*by_4+ay_3*bz_4+ay_4*bw_4 \\ az_1*bx_1+az_2*by_1+az_3*bz_1+az_4*bw_1 & az_1*bx_2+az_2*by_2+az_3*bz_2+az_4*bw_2 & az_1*bx_3+az_2*by_3+az_3*bz_3+az_4*bw_3 & az_1*bx_4+az_2*by_4+az_3*bz_4+az_4*bw_4 \\ av_1*bx_1+av_2*by_1+av_3*bz_1+av_4*bw_1 & av_1*bx_2+av_2*by_2+av_3*bz_2+av_4*bw_2 & av_1*bx_3+av_2*by_3+av_3*bz_3+av_4*bw_3 & av_1*bx_4+av_2*by_4+av_3*bz_4+av_4*bw_4 \end{bmatrix}$$

Figure 2.2: 4x4 Matrix multiplication

2.2. SYSTEM REQUIREMENT

In this assignment, you will analysis and design a 4x4 matrix multiplication system with the interface shown in figure 2.3

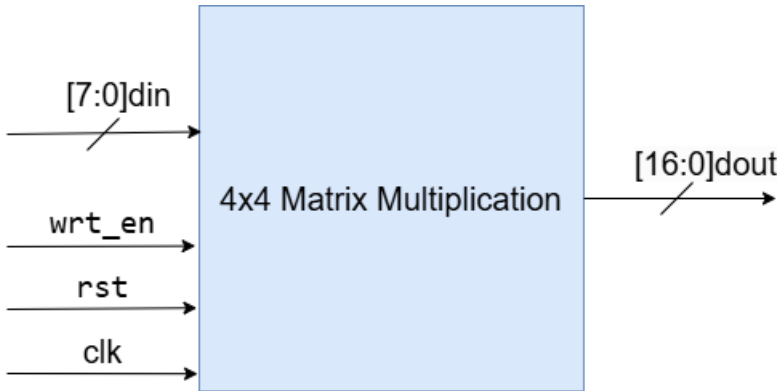


Figure 2.3: System block diagram

- **din**: The system will sequentially receive 32 signed integer input signals (The first 16 values of **matrix A** and the other of **matrix B**) with each signal having a size of 8 bits, in which MSB will be the sign bit and the remaining 7 bits will be the magnitude.
- **rst**: Synchronus reset input signal.
- **clk**: Clock input signal.
- **wrt_en**: Write enable input signal using for trigger writting input process.
- **dout**: The system will sequentially output 16 signed integer values of **matrix product** with each signal having a size of 17 bits, in which MSB will be the sign bit and the remaining 16 bits will be the magnitude.

Bonus: Write more modules and plan scenarios to demo on board.

If you have any questions or need help, don't hesitate to ask your instructor in class or via email.